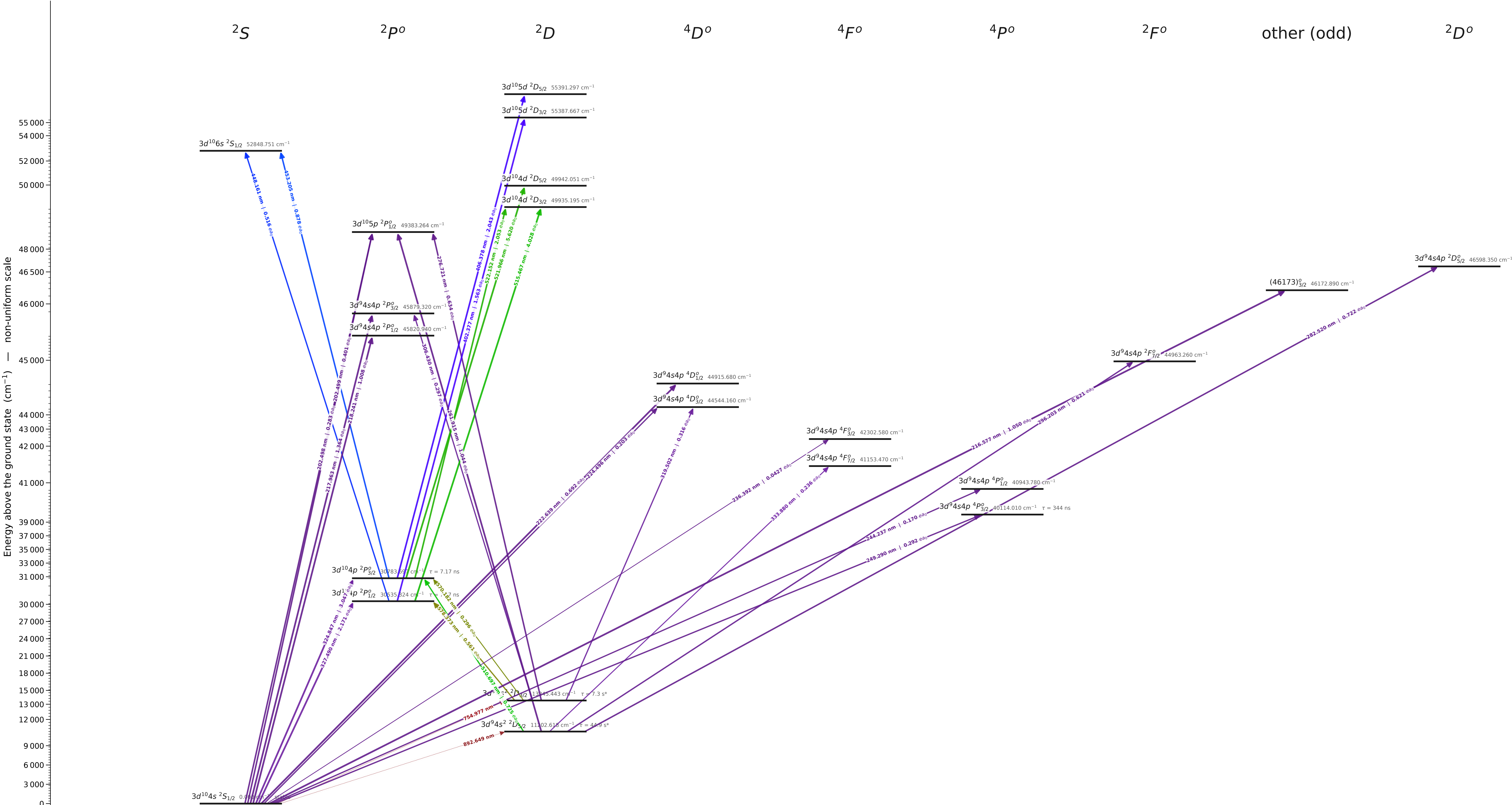
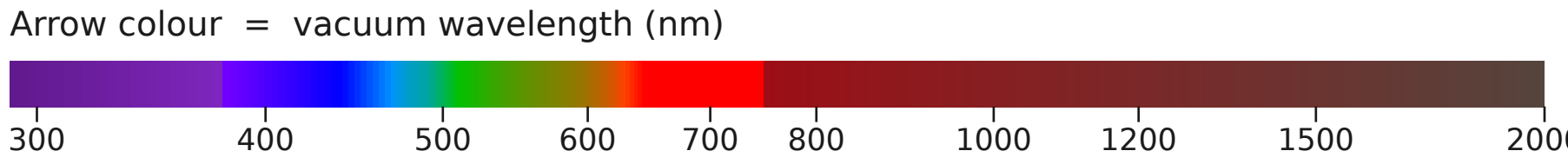


Copper

Cu (Cu I). Strongest classified transitions below 2 μm (NIST): 23 levels, 31 lines.

Cu⁺ ionisation limit 62317.46 cm⁻¹ = 7.72638 eV (160.469 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|(U||er||U')|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ (U er U') $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3d ¹⁰ 4s ² S _{1/2} - 3d ⁹ 4s4p ⁴ P ^o _{3/2}	249.2895	249.2143	1202.5878	0.292	2.790e+06	0.4627	96	249.2 nm line used for resonance-ionisation spectroscopy...
3d ¹⁰ 4s ² S _{1/2} - 3d ¹⁰ 4p ² P ^o _{3/2}	324.8473	324.7536	922.8720	3.047	1.372e+08	22.2	98.4	D2 resonance line 324.8 nm
3d ¹⁰ 4s ² S _{1/2} - 3d ¹⁰ 4p ² P ^o _{1/2}	327.4896	327.3952	915.4260	2.171	1.359e+08	21.89	98.8	D1 resonance line 327.4 nm
3d ¹⁰ 4s ² S _{1/2} - 3d ⁹ 4s ² ² D _{3/2}	754.9766	754.7688	397.0884	-	6.270e-07	2.18e-08	0	second proposed clock transition
3d ¹⁰ 4s ² S _{1/2} - 3d ⁹ 4s ² ² D _{5/2}	892.6485	892.4035	335.8460	-	2.230e-02	3.545e-09	100	E2 clock transition proposed for a Cu lattice clock

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁶³Cu hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3d ¹⁰ 4s ² S _{1/2}	5866.915(5)	-	F=1: -7333.64 F=2: +4400.19
3d ⁹ 4s ² ² D _{5/2}	749.100	186.000	F=1: -3802.57 F=2: -2453.18 F=3: -289.57 F=4: +2855.62
3d ⁹ 4s ² ² D _{3/2}	1851.000	137.400	F=0: -6769.50 F=1: -5055.90 F=2: -1491.30 F=3: +4199.10
3d ¹⁰ 4p ² P ^o _{3/2}	194.5(11)	-28.0(6)	F=0: -764.38 F=1: -541.88 F=2: -124.88 F=3: +430.62

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
3d ¹⁰ 4s ² S _{1/2} - 3d ⁹ 4s4p ⁴ P ^o _{3/2}	249.290	63-65	1055.4(26)

Shift = ν(first isotope) - ν(second isotope).